

Research Report

**MICROTUBULE IONIC CONDUCTION AND
ITS IMPLICATIONS FOR HIGHER
COGNITIVE FUNCTIONS**

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The neuronal cytoskeleton has been hypothesized to play a role in higher cognitive functions including learning, memory and consciousness. Experimental evidence suggests that both microtubules and actin filaments act as biological electrical wires that can transmit and amplify electric signals via the flow of condensed ion clouds. The potential transmission of electrical signals via the cytoskeleton is of extreme importance to the electrical activity of neurons in general. In this regard, the unique structure, geometry and electrostatics of microtubules are discussed with the expected impact on their specific functions within the neuron. Electric circuit models of ionic flow along microtubules are discussed in the context of experimental data, and the specific importance of both the tubulin C-terminal tail regions, and the nano-pore openings lining the microtubule wall is elucidated. Overall,

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these recent results suggest that ions, condensed around the surface of the major filaments of the cytoskeleton, flow along and through microtubules in the presence of potential differences, thus acting as transmission lines propagating intracellular signals in a given cell. The significance of this conductance to the functioning of the electrically active neuron, and to higher cognitive function is also discussed.

Keywords: Cytoskeleton; ionic conduction; cognitive function.

1. Introduction

The cytoskeleton is a web of filamentous protein polymers that define the structure of all eukaryotic cells. Composed of microtubules (MTs), actin filaments (AFs), and intermediate filaments (IFs), such as neurofilaments and keratin, the cytoskeleton provides architectural support for the cell, giving rise to its name, as well as facilitating cargo transport via motor proteins. Additionally, MTs perform specific functions relevant to cell motility, when composed as part of cilia or flagella, and to cell division, when forming the mitotic spindle. MTs have been discovered to play a role in the communication between the cell exterior and the cell nucleus, although the exact molecular mechanism of this process is not yet fully understood [1–3]. In recent years, it has been proposed that the AF and MT elements act as transmission lines to support the propagation of electrical signals within the cell [4–7].

Within axons and dendrites, the neuronal cytoskeleton displays a dense concentration of MTs stabilized by MT associated proteins (MAPs), specifically MAP-tau and MAP2, in polarized (axon), and mixed-polarity (dendrite) orientation. This is linked to the AF structures lining the walls of axons with specific MT architectures characterizing the presynaptic terminal [8]. In dendrites, the MT network connects to AF structures which enter into the postsynaptic density forming a highly elaborate framework. The IFs known as neurofilaments are the most abundant cytoskeletal components of large myelinated axons, although less is known about them compared to MTs and AFs. As such, due to the unique structure and function of the cytoskeleton in neurons, AFs and MTs play a major role in neuronal development, e.g., differentiation and synaptic generation, and neuronal plasticity.

The role of the neuronal cytoskeleton does not appear to be simply one of architecture or transport either. The functioning of the GABA_A receptor, a key target for the action of anesthetics, appears to rely on the microtubular cytoskeleton as its destruction has been shown to cause a 78% rundown in receptor currents [9]. Additionally, hypoxia-induced cytoskeletal re-ordering has been shown to be responsible for suppression of fast sodium channels in brainstem respiratory neurons [10], while drug-induced modifications of MTs altered the activity of calcium-dependent inactivation of calcium channels in isolated rat hippocampal neurons [11–14]. These results suggest that the neuronal cytoskeleton may mediate ion channel responses.

The specialized neuronal cytoskeletal architecture and function, and the inherently electric nature of neurons, point to the importance of the potential function of the cytoskeleton as a transmitter of electric signals. It is this potential electric

signaling function of the MT cytoskeleton that will be explored within the present article. Accordingly, the remainder of this article will be presented in the following manner: Sec. 2 discusses the involvement of the MT cytoskeleton in higher cognitive functions of learning, memory and consciousness, Sec. 3 will present the current experimental evidence for ionic conduction and theoretical models of these phenomena, Sec. 4 describes the link between cytoskeletal networks, the neuronal membrane, and neuron function, Sec. 5 discusses the particular importance of the C-terminal region and the enzymatic regulation of intracellular signaling, and finally Sec. 6 provides discussion and conclusions.

2. The Microtubule Cytoskeleton in Learning, Memory and Consciousness

While there is, as yet, no direct experimental confirmation that the role of the MT cytoskeleton includes electrical signal conduction in neurons, there is a large amount of experimental evidence supporting a role for MTs in higher cognitive function and by extension, neuronal signaling. The role of the neuronal cytoskeleton in learning and memory has been previously reviewed [4, 5, 15]. While in most tissues MTs exhibit growth, catastrophe (shrinkage) and tread-milling movement, they are less dynamic in neurons, largely because of the stabilizing effect of MAPs. However, rearrangement of the MT cytoskeleton does take place during learning. Learning has been shown to correlate with an increase in MTs [16], and to be impaired by the MT polymerization inhibitor colchicine [17–20], indicating that learning relies on dynamic MTs.

Correlations have been shown between learning and an increase in MAP2 [21], a MT cross-linking and stabilizing protein found mainly in dendrites [22, 23]. The converse is also true. Cerebral hypoperfusion, a condition which impairs learning, correlates with decreased MAP2 [24], and mice that exhibit learning and memory deficits also express less cortical MAP2 [25]. However, fear conditioning has been shown to produce significant changes in MAP2 immuno-histochemical staining [26–28], suggesting MAP2 breakdown during learning. This apparent contradiction is clarified when taken in the context of memory consolidation, the conversion of short-term memory into long-term memory. To reorganize the MT matrix in memory formation, the stabilizing protein MAP2 must first be broken down before reorganization can occur, and then reproduced to stabilize the new matrix. In fact, during memory consolidation, tubulin capping proteins increase their binding to other proteins [29], which is accompanied by MAP2 expression alterations [29, 30].

MAP-tau, another MT stabilizing protein found mainly in axons [22, 23], is also associated with learning and memory. Mice with increased levels of tau showed correlating improved memory before tau aggregation, and memory impairment after aggregation [31], and tau excess produced measurable impairment in memory in fruit fly and mice models [32, 33]. This suggests that an overabundance of tau causes excessive stabilization of the MT matrix inhibiting the dynamic reorganization

required in learning and memory. Indeed, a well-known hallmark of Alzheimer's disease, a disease of memory deterioration, is the accumulation of hyper-phosphorylated tau, unbound from MTs, into neurofibrillary tangles which results in neuronal degeneration and malformed neurites [34], suggesting that memory depends on proper functioning of the MT cytoskeleton.

Research on the molecular mechanisms of memory identifies Ca^{2+} /calmodulin-dependent protein kinase II (CaMKII) as crucial to long-term potentiation (LTP) contributing to learning and memory [35, 36]. Phosphorylation of MAP2 by CaMKII is fundamental to the potentiated synaptic response [37], and experiments indicate increased concentrations of dendritic mRNA for MAP2 and CaMKII during the intermediate phase of LTP [38]. CaMKII also directly phosphorylates the MT subunit protein tubulin on both its α and β monomer in the C-terminal region [39]. Additionally, learning via passive avoidance training induces changes in coexpressed MAP2 and protein kinase C (PKC) in rat models [40], and transgenic mice expressing altered MAP2 show decreased MAP2 phosphorylation by cAMP-dependent kinase (PKA) along with impaired contextual learning [41]. Such enzymatic events may be a triggering point for the MT matrix reorganization required for learning and memory formation.

While consciousness itself is hard to define, it is generally agreed upon that general anesthetics result in unconsciousness, making them a natural probe for functional sites of consciousness. Hallmarks of general anesthesia include hypnosis (unconsciousness) and amnesia (the prevention of memory formation), and it has been argued that hypnosis and amnesia are one and the same [42, 43]. The involvement of the cytoskeleton in memory has already been discussed above.

In recent years, evidence has accumulated for the interaction of anesthetics with neuronal MTs. Photo-labeling and gel electrophoresis have shown that the volatile anesthetic halothane binds directly to both α - and β -tubulin [44]. Differential in-gel electrophoresis investigations of protein expression determined that of 500 detectable proteins in rat cortical and hippocampal neurons, tubulin was altered prominently and was among the 2% of proteins affected by halothane, and among the 1% altered by isoflurane (being one of three proteins altered by both anesthetics) [45]. Additionally, tubulin has been found to be altered by desflurane [46], sevoflurane [47], as well as the neurosteroid anesthetics [48, 49].

The notion that the cytoskeleton plays a role in general anesthesia reaches back almost 40 years when Allison and Nunn first proposed the idea based on the depolymerization and structural alteration of MTs by volatile anesthetics [50]. This hypothesis was largely invalidated when it was found that the concentrations of anesthetics required to depolymerize MTs was far above the clinical dosage [100]. However, in more recent years this idea has been revived by quantum mobility theory [51, 52], which does not necessitate the depolymerization of MTs, but rather relies on hindrance of normal MT function by anesthetics. While the validity of quantum effects in MTs has been highly debated [53–55] this theory has resulted in much work being done in the area of cytoskeletal signaling and information processing in neurons.

Along the line of this argument, without the need to invoke quantum theory, hindrance of intracellular signaling, through the interaction of anesthetics with MT, could be a contributing factor to the mechanism of general anesthesia, and memory formation.

3. Ionic Signaling via Cytoskeletal Microtubules

The electric properties of biomolecules (and biomolecular structures composed of them), such as tubulin, the constituent protein of MTs, and actin, are of crucial importance to a biophysical understanding of cellular processes. This is all the more important to the understanding of the electrically active neuron which possesses an extremely complex cytoskeletal architecture not seen in the cells of other tissues [8].

MTs have been shown to be affected by external electric fields. It has been demonstrated that pulsed electric fields can align MTs assembled *in vitro* [56], and that AC electric fields can have a pronounced effect on the mitotic spindles in several cancerous cell lines [57], thus indicating the importance of the electrical properties of MTs. These electric properties may involve both intrinsic effects of the protein and ionic contributions from counterions attracted to and concentrated near the MT by its electric field. Unfortunately, conductance measurements on biopolymers are fraught with difficulties due to the structural heterogeneity and instability of these polymers, the liquid state of samples and the dependence of biological systems on environmental factors such as pH, temperature, and ionic concentrations. Nanotechnology advances have improved techniques allowing more precise investigations to take place. Fortunately, despite the inherent difficulties, a number of experimenters have investigated either the intrinsic [58–60] or ionic [61, 62] conductivities of MTs.

MTs are hollow cylindrical structures comprised of the globular protein tubulin. The tubulin molecule is a heterodimer composed of α - and β -tubulin monomers. The α - β complex that assembles into MTs has dimensions of approximately $8\text{ nm} \times 5\text{ nm} \times 4\text{ nm}$ [63] (see Fig. 1(a)). The inner and outer dimensions of an average MT cylinder are 15 nm and 25 nm, respectively, while their length typically varies from less than 200 nm to several micrometers, or more (see Fig. 1(b)). Each dimer possesses a highly flexible C-terminal tail region that can extend up to 4 nm from the dimer surface. Dimers bind together end-to-end (i.e., β to α) in string-like structures known as protofilaments. MTs in human cells typically have 13 protofilaments arranged side by side, generally in parallel arrangement [63], which wrap to form a cylinder, with the dimer C-terminal tails extending outwards from the MT surface. A slight shift between monomers in adjacent protofilaments results in a spiraling three-start left-handed helix of monomers. Two types of lattice structures have been observed [64]: the A-lattice structure possessing rotational symmetry and continuous wrapping, and the B-lattice which possesses a physical discontinuity called a seam running down the length of the MT (see Fig. 1). It is commonly accepted that, *in vivo*, MTs exhibit the B-lattice arrangement [65], however it has been suggested that dendritic MTs may prefer the A-lattice arrangement [52]. Due to the shape and spacing of

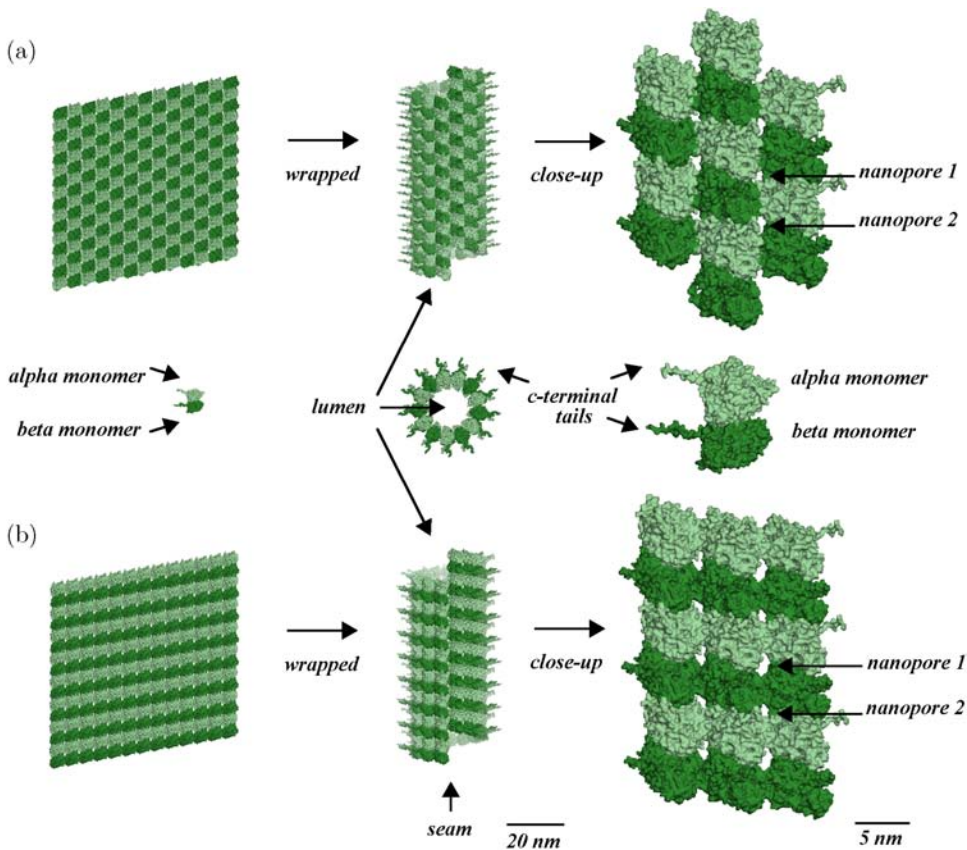


Fig. 1. Microtubule lattice geometries. (a) A-lattice showing perfect helical symmetry. (b) B-lattice showing the physical discontinuity known as a seam. Scale bar before close-up is 20 nm, after close-up is 5 nm.

the tubulin dimers the MT possesses small, nanometer size openings between the outer environment and the inner MT lumen (see Fig. 1). Two distinct types of these nanopores exist for each lattice geometry.

Ionic conductance along MTs relies both on their electrostatic and structural properties. In particular, the surface of the tubulin dimer is largely negative, with a specifically large negative charge located on the C-terminal tail, with the remainder concentrated along the protofilament ridges [66] (see Fig. 2). This results in an overall negatively charged surface for a MT with an average linear charge density of 52.2 electronic charges (e) per dimer, or approximately 85 e/nm [62]. In solution, positive ions are attracted to the MT surface, while negative ions are repelled, creating a condensed ion cloud to counter the MT negative surface charge (see Fig. 3(b)).

In the presence of an applied voltage, the loosely held counterion cloud is free to migrate, creating an ion flow and an associated electrical current, giving rise to ionic conduction (see Fig. 3(c)). Based upon Ohm's law, conductivity is proportional

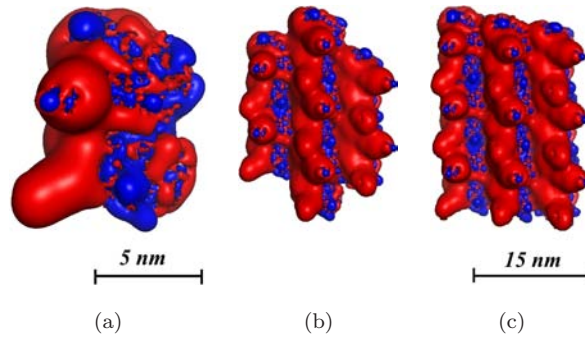


Fig. 2. Electrostatic map: Positive (blue) $+0.5 \text{ kT/e}$, Negative (red) -0.5 kT/e . (a) Tubulin dimer. (b) Patch of A-lattice microtubule showing a 7-dimer neighborhood. (c) Patch of B-lattice microtubule showing a 9-dimer neighborhood.

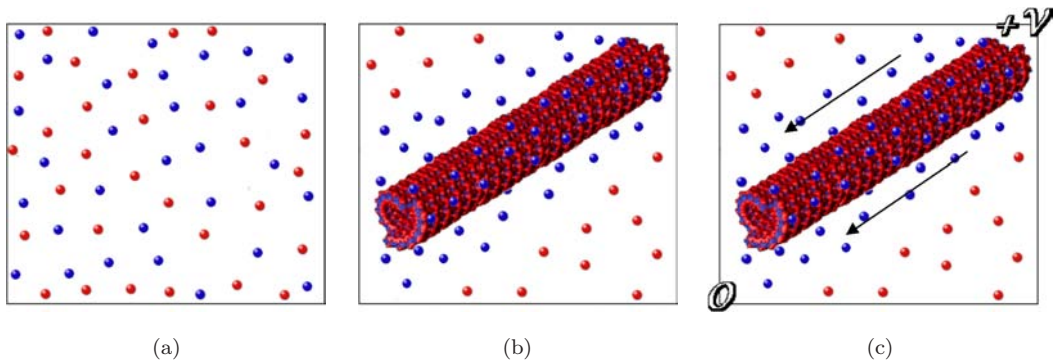


Fig. 3. Cartoon of ion flow along a microtubule. (a) Positive (blue) and negative (red) ions in solution. (b) Positive ions loosely condense around a microtubule due to its negative surface charge (red). (c) In the presence of a potential difference, positive ions flow down the microtubule. (C-terminal tails of the microtubule are not shown.)

to ionic concentration, and so the condensed ionic environments surrounding AFs and MTs are expected to be good conductors in the presence of a voltage source. Consequently, these biofilaments can be thought of as conductive elements in the cells' cytoskeletal circuitry.

In regards to the ionic conductivity, direct measurement attempts of conductivity along MTs in solution were made in filled microchannels and determined to have an upper bound of 90 S/m [61]. Results well within this bound were found by Minoura *et al.* using an electro-orientation method which yielded an MT conductivity of 150 mS/m for intact MTs and 90 mS/m for subtilisin-treated MTs [62]. The enzyme called subtilisin cleaves the C-terminal tail region from MTs. The difference in conductivity due to subtilisin treatment was shown to be roughly proportional to the change in charges per dimer associated with the absence of C-termini. This suggests that counterions on the surface, including those on the negatively charged C-termini, are responsible for the 150 mS/m observed conductivity via the flow of

these condensed ions around the surface of the MT, a value roughly 15 times larger than the conductivity of the surrounding solution (9.5 mS/m).

Priel *et al.* applied a patch-clamp technique to measure electrical currents flowing along MTs following electrical stimulation [7]. They showed that the electrical current generated in solution by a pulse of applied voltage across a 20–50 μm distance is approximately twice the value when MTs are present. Additionally, the group found a maximum MT conductance of 9 nS. These results taken together strongly suggest that, for MTs in solution, the flow of condensed ions around the MT surface provides the observed MT conductivity.

Ionic conductance around seamless, A-lattice MTs was modeled using a theory of resistor networks, where a single MT was represented as a two-dimensional periodic helical lattice of monomer pairs, represented as resistors creating a series of conducting pathways connected in parallel [67]. Furthermore, this model was tested against random walk simulations of ionic conduction along the monomer lattice by considering the relationship between the stochastic random walk process and the theory of electrical networks. Theoretical modeling and computer simulation yielded comparable results.

Another circuit-based investigation used a nonlinear cable-like model of a MT [68], involving resistive, capacitive, and inductive elements in series and in parallel. The model developed considered the MT lumen separated from the rest of the ions in the bulk solution by the counterion condensation cloud, which acts as a dielectric medium providing resistive and capacitive components to the circuit. Additionally, the helical structure of the MT provided an inductive component in analogy to the helical ion flow induced by a solenoid. Solitary wave solutions, of a potentially unstable nature, were predicted qualitatively supporting ionic signal propagation and amplification by MTs.

It must be noted that in the above models, the effect of MAPs was not considered, and neither were the effects of the tubulin C-terminal tails. The charge on the C-terminal tails accounts for approximately 40% of the total MT negative surface charge, with the exact charge being determined by the particular tubulin isotype (or a mixture thereof) comprising the MT. As such, it is expected that positive counterions will condense in larger concentrations around these tails to compensate for their large electrostatic negativity. Fluctuations in the highly flexible C-terminal tail domain must be associated with corresponding movements of the surrounding ion cloud to preserve charge compensation. Thus, the effects of C-terminal tail dynamics are expected to have a significant impact on MT ionic conduction.

Molecular dynamics modeling of the time-dependent conformational states of C-termini interacting with neighboring C-termini was carried out where the MT surface interacts with the C-termini via positively charged surface regions. The inclusion of interactions with adjacent MAPs indicated a potential for C-termini generated electrostatic perturbations of counterions to be transmitted between adjacent MTs via MAP2 [69, 70], forming the basis of a network of electrical elements created by

the architecture of the cytoskeleton in the cell. More recently, the effects of the C-terminal tails on ion transport along the MT have been specifically investigated [71]. In this case, ionic conductance was modeled in terms of positive ions hopping along a seamless lattice of ionic traps representing the large negative charge of the C-terminal tails along the MT surface. The simulations predicted critical ion concentrations required for conductance to emerge as a percolation threshold. It was shown that this effect can lead to the formation of complex conduction pathways, both of which are highly dependent on the charge of the C-terminal tails, something which is determined by the particular tubulin isoform whose distribution in the body is strongly tissue dependent with, for example, β III tubulin being largely concentrated in neurons.

A nonlinear RLC transmission line model of MTs viewed as rings of 13 tubulin dimers coupled together forming an electric circuit has been recently proposed [72]. In this model, the MT, with condensed counterions, was viewed as spatially separated from the rest of the ions in the bulk solution by a dielectric medium composed of an ionic layer depleted of positive ions. This layer provided both a fixed resistive component and a capacitive component that changes with the increasing concentration of counterions due to the flexibility of the C-terminal tail. Inductive components were also considered due to the 13 parallel currents arising from ionic wave flows along each protofilament. Solutions to this model indicated stable and robust anti-kink wave propagation with a velocity value close to experimental findings. It was also shown that even weak electric fields, such as a cellular intrinsic field, could tune both the direction and velocity of this propagation.

The latest research in this area generated an electric circuit model of the MT which incorporates not only the C-terminal tails, but also the nanopores within the MT wall, thus including not only ionic conductance along the outer surface of the MT, but conductance through the MT wall, and in the lumen as well [73]. This latest model effectively includes a 3D representation of the MT with ions flowing both along its surface and across it, in and out of the luminal space. Experimental evidence suggests importance for these nanopores in drug diffusion and MT-protein interactions. For example, the MT-stabilizing MAP tau, with its crucial function in axons, is believed to bind at a location in the MT lumen, suggesting that it passes through the nanopore to reach its binding site. It has been demonstrated that certain drugs (e.g., paclitaxel and epothilone) traverse the nanopores to find their binding sites in the MT lumen [74], or bind in the nanopore itself. Thus, it is assumed that based on their size, the nanopores are quite permeable to ions. Brownian dynamics simulations of ion conductance through B-lattice nanopores indicated a larger conductance traveling into the lumen for positive ions than for ions moving outside the MT, while negligible conductance was found for negative ions in either direction [73]. These results suggest that MTs give rise to complex electric conduction phenomena involving ions, solute molecules and potentially protons.

Finally, several authors have suggested that the flexible C-terminal tails can interact with positively charged residues on the body of tubulin [75, 76]. The formation and disruption of these interactions was proposed to act as a stochastic alternating voltage source on the cations close to outer surface of MT. Simulations including the effects of the asymmetrical conductance and this internal voltage source indicated that an ionic current can travel through the MT wall and propagate down the MT lumen [73]. This current adds directly to the longitudinal current along the outer MT surface arising from an external voltage source, indicating that MT can amplify low-intensity endogenous currents within the cell. Because the voltage source arising from the dynamic behavior of C-terminal tails drives cations into the lumen of the MT all along its outer surface, this effect could be an important factor enabling the dissipation of a potential gradient to be focused towards one end of the MT, regardless of the orientation of the source relative to the MT.

Overall, these results suggest that counterions condensed around the surface of the MT cytoskeleton are allowed to flow in the presence of potential differences, effectively making it a network of biological electrical wires. This may have important implications in the coupling of intracellular signals in all cells. However, as stated earlier, electrically active cells, such as neurons, generate potential differences via the flux of charged ions, and in these instances the mechanism of conduction along MT may be of extreme significance.

4. Connections Between Microtubule and Actin Networks and the Neuronal Membrane

Both AFs and MTs form networks using various binding proteins. There are over 100 actin-binding proteins implicated in cell types and locations. Of particular relevance are α -actinin and spectrin that promote neural plasticity. The protein α -actinin is known to link AFs to membrane channels, thereby coupling channel activity to the intracellular network. MAPs in neurons are responsible for the structured parallel network observed in the axon, as well as the mixed-polarity arrangement in dendrites. Specifically, MAP-tau is abundant in the axon, creating polarized bundles of MTs to span its length, while in developed neurons MAP2 is present in dendrites where it stabilizes the network of MTs to form a mixed polarity structure. Since MAP2 is relatively heavy and long compared to MAP-tau, the average distance between the MTs in the network developed in the axon is about 20 nm whereas in dendrites it measures 50–90 nm.

The dynamic interactions between both AF and MT cytoskeletons are central to neuronal cell morphology and function. Only recently have clear functional interactions between those cytoskeletal structures become apparent. Association of MT with AF in filopodia appears to guide MT growth along the most efficient path and play a key role in neurite initiation [77]. AF-MT interactions are also important for growth cone turning underlying axonal path-finding mechanisms [78].

Like MTs, AFs are also highly charged polymers, with a negative charge density of about $4e/nm$ [79]. They are thinner than MTs with diameters of about 7 nm, they are less rigid by a factor of about 300, are typically shorter, often cross-linked by other proteins into bundles, and are closely associated with neuronal cell membranes [80]. Application of a patch-clamp technique to measure electrical currents flowing along AFs following electrical stimulation found that the charge transferred was consistent with the dynamic behavior of ion flow along the AF longitudinal filament axis and that this flow was doubled by including AFs in the solution [81]. Additionally, osmotic stress applied to a polymerized actin solution revealed a reduction in actin surface charge density with increasing stress and a nonlinear electric response upon application of electric fields, indicating that F-actin in solution may display non-ideal electro-osmotic properties consistent with ionic “cable” behavior [82]. Also, measured electrical signals from the end of an AF after an input voltage was applied indicate that AFs support ionic waves in the form of axial nonlinear currents [79]. Thus, like MTs, AFs are capable of ionic signal conduction and amplification.

In neurons, there is an increased presence of binding proteins capable of interacting with both MTs and F-actin, mediating signaling between both cytoskeletons. For example, MAP2 binds to F-actin efficiently, and induces actin bundle formation. MAP2 has also been found to localize in regions devoid of MTs in neurons [83], in particular in dendritic spines. Since actin bundles enter the spine neck and bind to the postsynaptic density (PSD), and on the other end are connected to the MT network filling the dendritic shaft, it has been hypothesized that a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and helps modify the electrical response of the neuron. In this picture, MTs arranged in networks of mixed polarity, receive signals in the form of electrical perturbations, from synapses via AFs connected to MTs by MAP2 [84], or via direct MT connections to PSD proteins by molecules such as CRIPT [85]. The MT network operates on the propagating waves and modulates the information it carries to control the output measured by remote membrane areas. This intracellular information processing mechanism effectively regulates the cell’s electrical response [5].

Since voltage-gated Na^+ and Ca^{2+} channels play an important role in amplifying the excitatory postsynaptic potential (EPSP) responses towards the soma, channel regulation will be important also for dendritic interactions in the vicinity of the synaptic input. Voltage-gated channels may dynamically modify the local input properties — resistance and time constant, which in turn would influence the spatio-temporal integration of EPSPs and inhibitory postsynaptic potentials (IPSPs), thus, promoting highly nonlinear interactions. Hence, the nonlinear properties of dendrites may act as amplification/modulation devices. This control of membrane potential would modify synaptic integration and therefore modulate the whole neural response.

5. Enzymatic Control of Signal Propagation via Modification of Tubulin C-Termini

Reviews of the specialized role of tubulin isoforms suggest that the evolution of new cell types corresponds to the origin of specific tubulin isoforms [86]. It is interesting to note that of all tissues the brain possesses the largest variety of tubulin isotypes [87]. In particular, this suggests high probability for a very significant role of tubulin isoforms in neural network complexity and its origin [88].

The C-terminal is the most variable region among the many different tissue specific isotypes of tubulin, and, as such, has been suggested to mediate the specific cellular functions of MTs [87, 89]. The importance of the C-termini to the structure and function of MTs has been reviewed extensively [90–92], and includes effects on MT polymerization, and interaction with MAPs and a host of other proteins.

In this regard, it has been suggested that this C-terminal region acts as a chaperone to interactions with other proteins such as MAPs [93]. In fact, it has been shown that MAP tau triggers structural changes in elements on the MT surface [94], and simulations indicate these elements may be the C-termini [90]. The C-termini are also the sites of many of the post-translational modifications of tubulin, including polyglutamylation, polyglycylation, detyrosination/tyrosination, removal of glutamic acid and phosphorylation of serine and tyrosine residues [95, 96]. These modifications are the result of enzymatic processes [97], and considering the involvement of the C-termini in MT polymerization and interaction with MAPs, are expected to have dramatic dynamic effects on the MT network. For example, modifications that change the electrostatic nature of the C-termini, by adding or removing charged residues, would effect the capacitance of the condensed ionic atmosphere surrounding a MT, as well as the magnitude of the voltage source arising from tail–body interactions. Thus, should the C-termini play a significant role in ionic conduction along MTs, as proposed by the simulations mentioned above, then these enzyme-triggered events would have great effects on ion flow, providing a mechanism for the regulation of intracellular signaling.

6. Discussion and Conclusions

MTs composed of tubulin and some associated proteins are ubiquitous cellular structures of eukaryotic cells important for organizing cytoplasmic properties and cellular movement. Their α - and β -tubulin subunits constitute heterodimers that associate into protofilaments with a strict α/β alternation. Protofilaments are able to form rings and spirals or associate laterally building up both different kinds of sheets and MTs or even microtubules, depending on the prevailing environmental conditions during the assembly process. The various geometrical possibilities that MTs, AFs and other proteins present in the cytoskeleton give rise to the design of very complex architectures of these filamentous structural elements of neurons. However, their intrinsic electrical conduction and dielectric polarization properties open

an additional, entirely new spectrum of functional applications within the cellular machinery.

Due to the presence of a large electric dipole in each of the tubulin units (including the layer of bound structured water surrounding the solvent exposed area) and the negatively charged tubulin C-termini, electric fields are expected both to transport suspended MTs into a desired direction and to steer MTs actively driven by motor proteins. Correspondingly, in experiments performed at nearly physiological conditions, (pH 6.8 and about 120 mM ionic strength) individual suspended MTs were found to move steadily towards the positive electrode with an electrophoretic mobility: $2.6 \times 10^{-4} \text{ cm}^2/\text{Vs}$ [98]. The electrophoretic mobility decreases with ionic strength and pH. The MT movement changed direction to the anode at pH 4.1, indicating a positive net charge has been created on MTs when they were kept at this low value of pH. Interpolation of the isoelectric point gives a value of about pH 4.2 for MTs, which is remarkably lower than that of α - and β -tubulin monomers found to be near pH 5.5. At pH 6.8, external electric fields forced MTs to glide across a kinesin-coated surface into a preferred direction to the anode, in accordance with the negative net charge of the MTs at this pH.

Interestingly for potential nanotechnology applications, the surface of tubulin molecules exposes a defined pattern of amino acid residues (histidines) that provides active sites for nucleation, organization, and binding of metal nano-particles. The doping or decorating of the MT with these nano-particles can be performed by using various processes such as precipitation, solvent evaporation, and vapor phase reactions. The formation of nanometre-size (diameter: 1.5–3.5 nm) immobilized noble metal particles on the MT surface has been experimentally demonstrated [99]. The particles formed have shown definite patterns reflecting the regular arrangement of the α - and β -tubulin subunits within the MT. By further particle growth, a quasi-continuous MT coating can be obtained resulting in Pd nanowires and, furthermore these wires can be assembled into circuits [99] (see Fig. 4). Such an isomolecular organization of nano-particles into functional 2D and 3D structures is a potential method for developing electronic devices with novel biophysical properties. These nano-particles may be deposited randomly or pseudo-randomly within the underlying MT geometry. This results in the formation of a MT with mechanical, electrical and magnetic properties that are different from the original MT. Electrically modified MTs can become useful in applications such as artificial nerves, neural networks and logic circuits, biosensors, neuron-to-neuron couplers, bioisolators and bioswitches. It is feasible to apply the intrinsic properties of MTs, as well as electrically and magnetically modified MTs, to enable the construction of electronic devices and electronic circuits. For example, a four-terminal device can be fabricated from MTs by depositing two parallel gold pads on a substrate, and adding a layer of MAPs between the gold pads to aid in the contact between the gold pads and the MTs. The application of an electric field across the two gold pads will align the MTs in a parallel plane, and allow for a low resistance state. Conversely, the removal or lack of an electric field between the gold pads provides for random

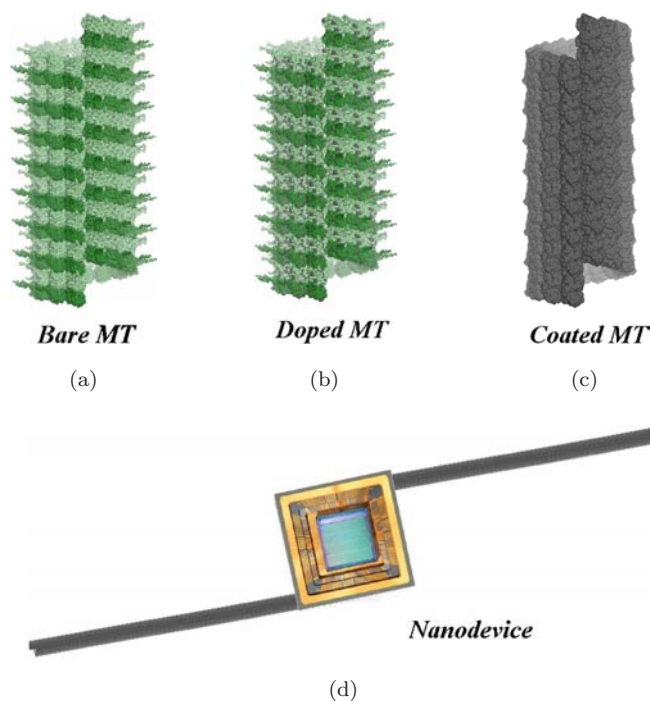


Fig. 4. Nanowires. (a) Bare B-lattice microtubule. (b) B-lattice microtubule doped with silver nano-particles at histidine residues. (c) Silver nano-particle-coated microtubule. (d) Hypothetical nano-device wired with silver coated microtubules.

orientation of the MTs with respect to each other and to the gold pads. Variations on the configuration of the gold pads provide for inter-changeability between high- and low-resistance states when an electric field is applied. The above four-terminal logic device may also be used to provide electrical isolation of unwanted signals in the presence of an electric field. Noble metals that were used to decorate or coat the MTs for electric field response include gold and palladium. The use of ferromagnetic materials such as iron to decorate MTs may lead to creation of bioengineered structures that are highly responsive to an applied magnetic field.

The conventional thinking ascribes consciousness to the central nervous system only and hence, it is crucial to investigate what makes the MTs there suitable for information processing or at least for aiding the cognitive process. This would be in contrast to the MTs formed in almost every other cell of the body. On the other hand, cell shape determines whether or not external electric fields, such as that of the nerve impulse, may penetrate the cell. This implies a feedback mechanism between MTs and the cell shape. The goal of the studies reviewed in this article is to learn more about the function of tubulin and MTs and to discover novel properties which make them particularly useful within the nervous system. We hope to have shed light on the reason why MTs are important for neuron function and consciousness. We have also concluded that modifying biophysical properties of MTs using nanotechnology

may open a new range of possibilities such as bioelectronic artificial intelligence devices.

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